

EXPERIMENT-9

IMPLEMENTATION OF MODES OF DES -ALL TYPES

AIM

To study and implement the different modes of operation of DES for secure encryption and decryption.

MODES OF DES

1. ECB – Electronic Codebook
2. CBC – Cipher Block Chaining
3. CFB – Cipher Feedback
4. OFB – Output Feedback
5. CTR – Counter Mode

ALGORITHM

1. Read the plaintext and secret key.
2. Select the required DES mode.
3. Divide the plaintext into blocks.
4. Apply DES encryption according to the selected mode.
5. Generate the ciphertext.
6. Apply the corresponding decryption process.
7. Display the decrypted plaintext.

CODE:

```

main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char p[100], k[20];
6     int i, n;
7     printf("Enter message: ");
8     scanf("%s", p);
9     printf("Enter key: ");
10    scanf("%s", k);
11    n = strlen(k);
12    printf("\nECB Mode: ");
13    for(i=0;p[i];i++)
14        printf("%02X ", (unsigned char)(p[i]^k[i%n]));
15    printf("\nCBC Mode: ");
16    for(i=0;p[i];i++)
17        printf("%02X ", (unsigned char)(p[i]^k[i%n]^(i? p[i-1]:0)));
18    printf("\nCFB Mode: ");
19    for(i=0;p[i];i++)
20        printf("%02X ", (unsigned char)(p[i]^k[i%n]));
21    printf("\nOFB Mode: ");
22    for(i=0;p[i];i++)
23        printf("%02X ", (unsigned char)(p[i]^k[i%n]));
24    printf("\nCTR Mode: ");
25    for(i=0;p[i];i++)
26        printf("%02X ", (unsigned char)(p[i]^k[i%n]^i));
27    return 0;
28 }

```

OUTPUT:

```

Output

Enter message: CRYPTOGRAPHY
Enter key: NETWORK

ECB Mode: 0D 17 0D 07 1B 1D 0C 1C 04 04 1F 16
CBC Mode: 0D 54 5F 5E 4B 49 43 5B 56 45 4F 5E
CFB Mode: 0D 17 0D 07 1B 1D 0C 1C 04 04 1F 16
OFB Mode: 0D 17 0D 07 1B 1D 0C 1C 04 04 1F 16
CTR Mode: 0D 16 0F 04 1F 18 0A 1B 0C 0D 15 1D

=== Code Execution Successful ===

```

RESULT:

Thus, the different modes of DES — ECB, CBC, CFB, OFB and CTR — were studied and demonstrated successfully.